Methods and Principles of Designing Modern Secure Systems

Lecture 5

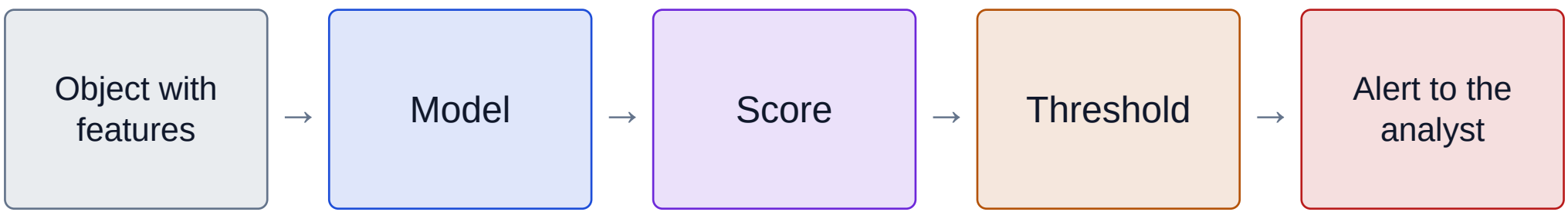
Classification Models for Incident Detection

Class imbalance · metrics · threshold · interpretation

Lecture outline

1. The model’s role in detection
2. Class imbalance
3. Classification models
4. Metrics and threshold choice
5. Model interpretation
6. Operation and model card

From object to alert



Quality is set by two decisions: which model and where the threshold is

1

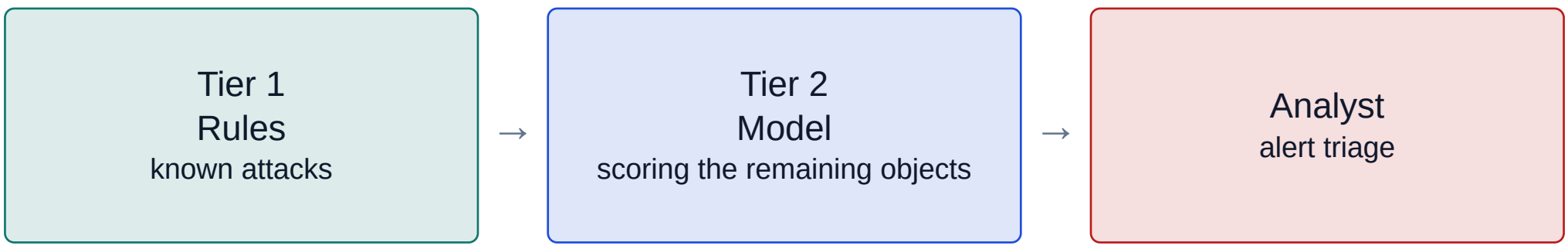
The model’s role in detection

when a model is needed rather than a rule

Rule or model

Problem condition	Rule	Model
The attack is described by one threshold condition	Suitable	Overkill
A combination of many weak features matters	Hard to write	Suitable
No labeled attack examples	Suitable	Cannot be trained
Every decision must be justified	Suitable	Interpretable model
No monitoring and retraining	Suitable	Quality degrades

Three tiers



The model complements rules, it does not replace them

Training dataset

4,116

training objects · 15 attacks

1,215

validation · 6 attacks

1,217

test · 6 attacks

Object: a source address over five minutes

Features: attempts, failure share, accounts, night

Time-based split; days 35 and 46 are gaps

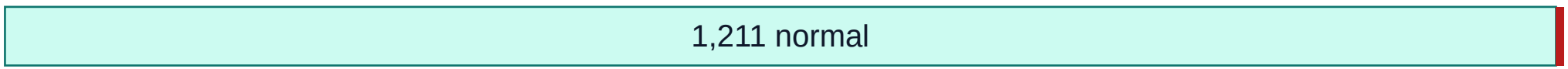
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Class imbalance

few attacks, a great deal of normal traffic

Accuracy under imbalance

Test set: 1,217 objects



6 attacks · 0.49 %

Metric	Always “normal”	Logistic regression
Attacks found	0 of 6	**4 of 6**
Accuracy	**0.995**	0.987
Recall	0.000	**0.667**
Precision	—	**0.222**
F1	0.000	**0.333**
Balanced Accuracy	0.500	**0.828**

By Accuracy, the winner is a model that finds no attacks at all

Methods for handling imbalance

Undersampling normal	Oversampling attacks	Class weights	Threshold
keeps real data, loses information	copies or synthetic data; no new attack types	an error on an attack weighs more; data unchanged	chosen on the validation set instead of 0.5

Recommended order

1 · Class weights → 2 · Threshold on the validation set → 3 · Resampling the training set

Resample only the training set; the score of a weighted model is not a probability

3

Classification models

which model to choose and how to compare

Logistic regression and ensembles

Logistic regression	Random forest	Gradient boosting	Neural network
weighted sum of features; baseline model	independent trees, averaging	each tree corrects the errors of the previous ones	for sequences and very large volumes

Model comparison

Model	Training	Explainability	When to use
Logistic regression	Very fast	High: weights	Baseline model
Random forest	Fast	Medium	Non-linear combinations
Gradient boosting	Medium	Medium: SHAP	Default choice for tabular data
Neural network	Slow	Low	Sequences, text

Three models on the training dataset

Model	AP, validation	AP, test	ROC-AUC, validation	ROC-AUC, test
Logistic regression	0.156	0.118	**0.882**	**0.883**
Random forest	**0.503**	**0.337**	0.754	0.622
Gradient boosting	0.337	0.005	0.656	0.430

AP is average precision; for a random guess it equals the attack share, about 0.005

Average precision across time blocks

Block	Days 1–20 → 22–28	Days 1–27 → 29–35	Days 1–34 → 36–45
Attacks in validation	2	4	6
Logistic regression	0.502	0.277	0.156
Random forest	0.502	0.752	0.503
Gradient boosting	0.502	0.752	0.337

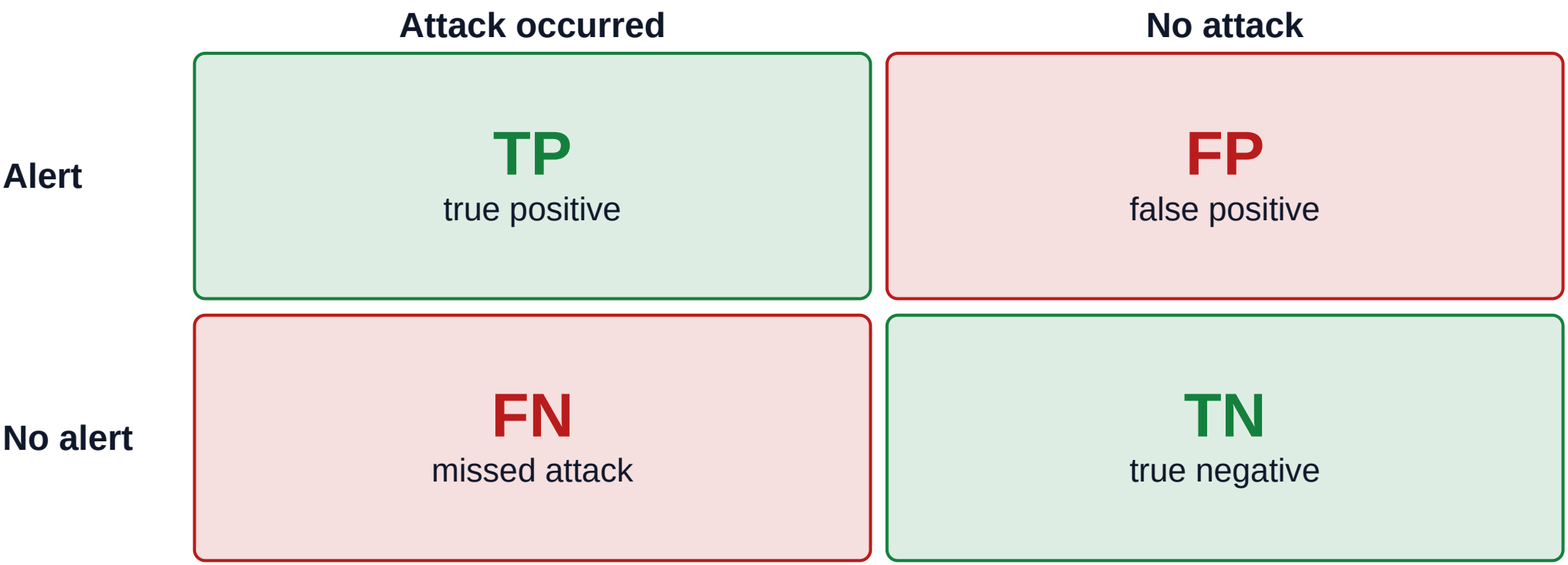
The model ranking changes; with few attacks, evaluate across several blocks

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Metrics and threshold choice

how to evaluate a model and where to draw the line

Four outcomes



Precision and recall

Precision

$$TP / (TP + FP)$$

what share of alerts are attacks

Recall

$$TP / (TP + FN)$$

what share of attacks is found

Higher threshold — fewer false alerts, but more missed attacks

F-score: β sets the weight of recall

$$F_\beta = \frac{(1+\beta^2)\cdot \text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}}$$

β = 1 · F1
precision and recall weigh equally

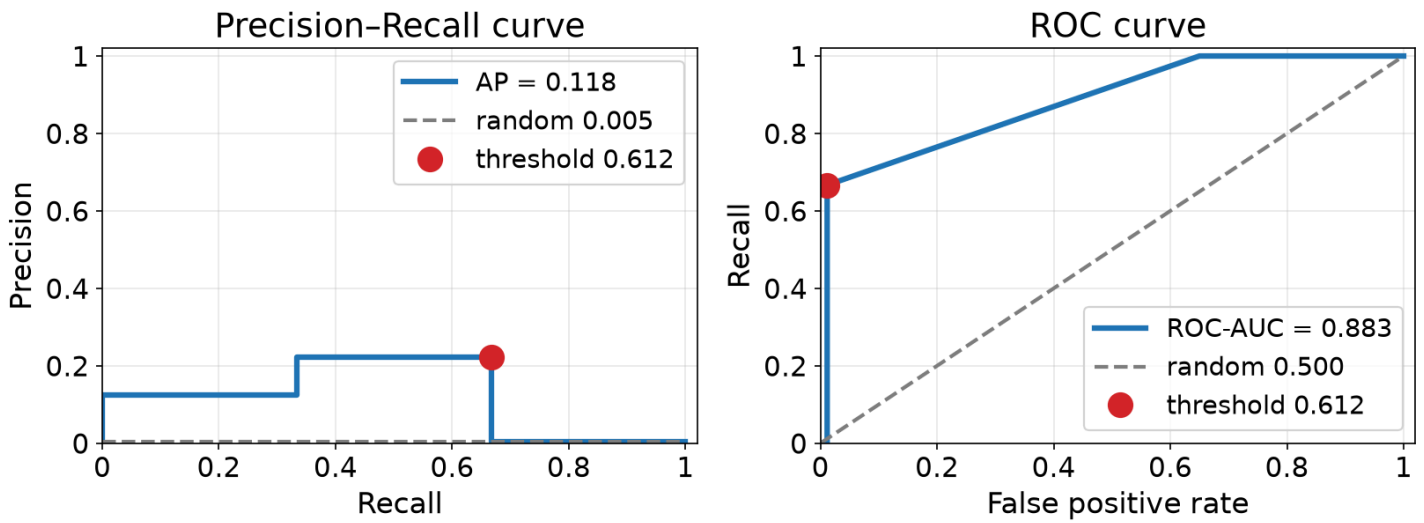
β = 2 · F2
recall matters more: a miss is costlier

β = 0.5 · F0.5
precision matters more: false alerts are costly

Illustrative example	Precision	Recall	F1	F2	F0.5
A: few false alarms, many misses	0.9	0.3	0.450	0.346	**0.643**
B: few misses, many false alarms	0.3	0.9	0.450	**0.643**	0.346

F1 is the same, but F2 picks model B and F0.5 picks model A

Precision–recall curve and ROC



AP = 0.118
random level 0.005

ROC-AUC = 0.883
looks fine

Threshold 0.612
precision 0.222 · recall 0.667

A curve shows all thresholds at once; with rare attacks ROC-AUC overstates quality

Threshold: workload and attacks detected

Validation set · ten days · six attacks

Threshold	Alerts per day	Attacks in the set (max TP)	Attacks detected (TP)
0.3	**81**	6	**6**
0.5	1.7	6	4
0.7	1.6	6	3
0.8	1.2	6	2
0.99	1.1	6	1

Lower threshold — more alerts and more finds · on the test set no attacks are detected at 0.8 and above

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Model interpretation

why the model decided so

Logistic regression weights

Feature	Weight	How to read it
Number of attempts	+0.953	more attempts — higher score
Failure share	−0.614	all else equal — lower score
Number of accounts	−0.311	all else equal — lower score
Night hour	**−2.326**	night windows get a low score

All attacks in the data happen in daytime: the model took this as a pattern

Permutation feature importance

1 · Model quality
AP = 0.156

→

2 · Shuffle the values of one feature

→

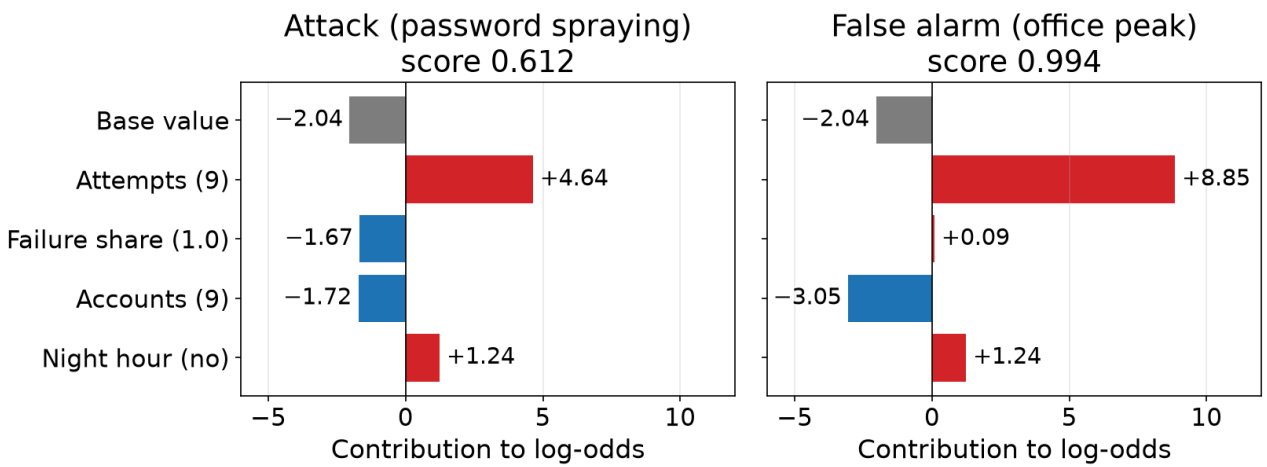
3 · Importance = quality before − quality after

Feature	AP after	Importance	Conclusion
Number of attempts	0.006	**0.150**	relied on: quality drops to random
Number of accounts	0.123	0.033	weak influence
Night hour	0.138	0.018 ± 0.052	indistinguishable from zero
Failure share	**0.312**	**−0.156**	harmful: quality went up

An important feature: quality drops after shuffling

SHAP values: what the score is made of

Base value + feature contributions = model output · red raise the score, blue lower it



Attack: the number of attempts raises the score, failures and accounts lower it

Office peak: the number of attempts (+8.85) drowns out everything else

Permutation vs SHAP: the difference

Permutation
“How much does the model suffer without this feature?”

SHAP
“What exactly did this feature do to this object’s score?”

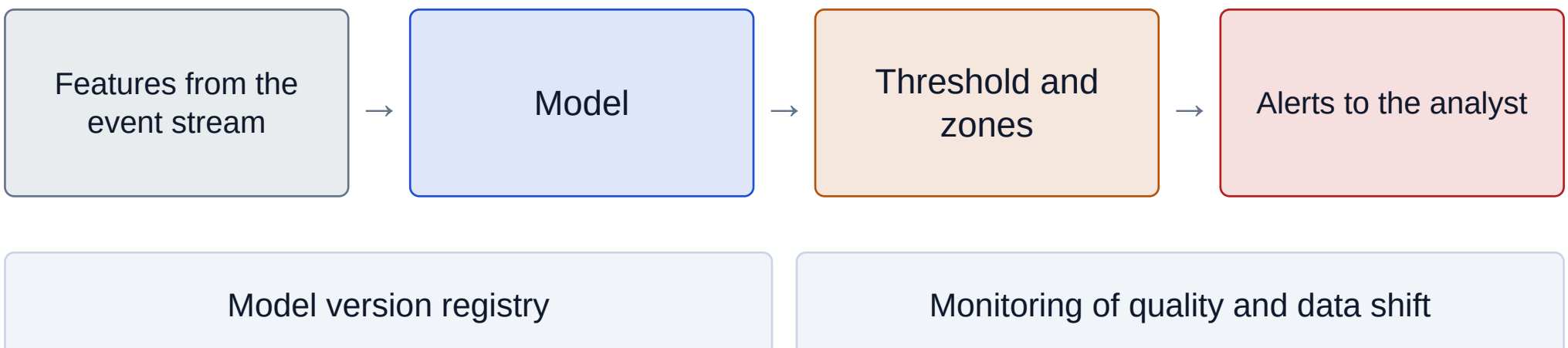
Property	Permutation	SHAP
What it does	Shuffles feature values, measures the drop in quality	Decomposes the model output into feature contributions
Computed for	The whole set: one number per feature	Each object separately
What it shows	Usefulness of a feature for quality (AP)	Influence of a feature on the model output
Needs labels?	Yes: quality cannot be computed without them	No: the model and data are enough
Sign	Drop — important; about 0 — unused; below 0 — harmful	Plus raises the score, minus lowers it

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Operation and model card

from a model to a system

The model in a monitoring system



Operational requirements

Reproducibility

data, versions, random seeds

Versioning

model, data and features tied to the threshold

Latency

fits within response deadlines

Monitoring

quality and input data shift

Rollback

fast return to the previous version

Protection

restricted access to the model and threshold

Scoring modes

Mode	When used	Example
Batch	latency is acceptable, large volume	password guessing by windows
Near real time	fast reaction needed	connection to a C2 server
Real time	critical systems, low throughput	embedded sensor

Model card

Details

algorithm, version, date, author

Intended use

what it is and is not for

Training data

source, period, attack share

Evaluation data

subset boundaries

Metrics

AP, precision, recall

Threshold

value and rationale

Limitations

attack types, degradation, retraining

Model card example

Section	Content
Model	Logistic regression, class weights, four features
Intended use	Scoring how suspicious a source is over 5 minutes from the login log
Training	Days 1–34: 4,116 objects, 15 attacks
Evaluation	Validation: days 36–45, test: days 47–56, 6 attacks each
Metrics, test	AP 0.118; precision 0.222; recall 0.667; F1 0.333
Threshold	0.612: by F2 on validation; coincided with the cost minimum
Limitations	Few attacks; misses a single login after guessing; “no attacks at night”

Key points

1 The model’s role	The model complements rules
2 Imbalance	Accuracy is uninformative when attacks are rare
3 Models	Baseline and ensembles; few attacks — unstable conclusions
4 Metrics and threshold	Average precision; threshold from validation
5 Interpretation	Weights, permutation, SHAP — hunting for errors
6 Operation	Reproducibility, monitoring, model card

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Classification Models for Incident Detection

Class imbalance · metrics · threshold · interpretation